

```
In [ ]: # --- Environment Setup ---
import os
import sys
# Support both Google Colab and Local repository paths
colab_dir = "/content/drive/MyDrive/algo_trading_project"
if os.path.exists(colab_dir):
    project_root = colab_dir
else:
    project_root = os.path.abspath("..") if os.path.exists("../strategies") else

strategies_path = os.path.join(project_root, "strategies")
if strategies_path not in sys.path:
    sys.path.append(strategies_path)
# -----
```

NIFTY50 MACD + RSI Strategy – Project Summary

This notebook summarises the NIFTY50 trading strategy project, from data collection to backtest evaluation, using a long/short **MACD + RSI** rule.

Objective

To test how a simple MACD + RSI rule performs on 20 large-cap NIFTY50 stocks (Jan 2018 – Dec 2023) and to evaluate it with return, Sharpe ratio, drawdown and a buy-and-hold benchmark.

Project Structure

- **01_data_collection.ipynb**: Fetches historical stock data for 20 NIFTY50 stocks from Yahoo Finance.
- **02_feature_engineering.ipynb**: Computes technical indicators (SMA-20, EMA-20, MACD, RSI, Bollinger Bands, ATR).
- **03_strategy_backtesting.ipynb**: Runs the MACD + RSI backtest and computes performance metrics.
- **04_strategy_visualization.ipynb**: Visualizes performance of the top/bottom stocks.
- **05_project_summary.ipynb**: (this notebook) summarises the project.
- **06_ml_signal_classifier.ipynb**: Tests whether technical features predict 5-day forward returns.

All results are saved in `results/strategy_results.csv`. Signals are lagged one day (no look-ahead). `Total Return = Final Value - 1`.

```
In [ ]: if 'project_root' not in globals():
import os
project_root = os.path.abspath("..") if os.path.exists("../strategies") else

import pandas as pd

# Load results
results = pd.read_csv(f"{project_root}/results/strategy_results.csv")
results.sort_values("Total Return", ascending=False).head(10)
```



Top Performing Stocks (by Total Return)

```
In [ ]: import matplotlib.pyplot as plt

top_stocks = results.sort_values("Total Return", ascending=False).head(5)

plt.figure(figsize=(10, 5))
plt.bar(top_stocks["Ticker"], top_stocks["Total Return"], color="green")
plt.title("Top 5 Stocks by Strategy Return")
plt.ylabel("Total Return")
plt.grid(True)
plt.show()
```

▼ Worst Performing Stocks

```
In [ ]: worst_stocks = results.sort_values("Total Return").head(5)

plt.figure(figsize=(10, 5))
plt.bar(worst_stocks["Ticker"], worst_stocks["Total Return"], color="red")
plt.title("Bottom 5 Stocks by Strategy Return")
plt.ylabel("Total Return")
plt.grid(True)
plt.show()
```



Strategy Highlights

- Only **2 of 20** stocks ended with a positive return (`BHARTIARTL.NS` +184%, `SBIN.NS` +178%); 2 never traded; **16 lost money**.
- The rule fires rarely: **0–2 position changes per stock** over six years, so each result is essentially one directional bet.
- `BHARTIARTL.NS` had 0 position changes and its return equals buy-and-hold exactly: that result is not evidence of a working strategy.
- The strategy beat buy-and-hold on **1 of 20** stocks (`SBIN.NS`) and lost to it on 18.
- Simple average across the 20 stocks: strategy about –22% vs buy-and-hold about +180%.
- Worst result: `BAJFINANCE.NS` (–86%).

ML Experiment: Can Technical Features Predict 5-Day Returns?

An earlier version of this notebook labelled a "Buy" using MACD and RSI, then used MACD and RSI as features. That is target leakage, and it produced a meaningless AUC of 1.00 (only 3 positive cases in the test set). It has been replaced.

Corrected setup

- **Target:** 1 if the stock's return over the next 5 trading days is positive.
- **Features:** scale-free ratios of MACD, RSI, SMA/EMA distance, volatility, momentum and volume, computed per ticker.
- **Split:** chronological (first 80% of dates for training, last 20% for testing, with a 10-day gap).
- **Model:** RandomForestClassifier (300 trees, `min_samples_leaf=50`, balanced class weights).

Result (out of sample)

- ROC AUC: **0.529**, close to random (0.5).
- Accuracy: 0.51, below the 0.561 share of up-moves in the test set.

Conclusion: these technical features show no reliable predictive edge on this data.



Limitations & Next Steps

- No transaction costs or slippage; short positions are assumed frictionless.
- The 20 stocks were hand-picked from today's large caps, which introduces survivorship and selection bias.
- The rule trades too rarely to judge; test signals that fire more often and add an equal-weight portfolio-level backtest.
- Use historical index membership and walk-forward validation for any further ML work.



Conclusion

A simple MACD + RSI rule did not beat buy-and-hold on 19 of 20 stocks. The project's main value is the workflow: careful backtesting, a buy-and-hold benchmark, and catching leakage in the first ML attempt.